

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'A packet filter firewall filters at the' is 'Network layer'.
2. True or False: The correct answer to '_____ is also a part of darknet that is employed for transferring files anonymously.' is 'Tor'.
3. True or False: The correct answer to 'Which of the following is an authentication method?' is 'None of the above'.
4. True or False: The correct answer to 'A _____ is a sequential segment of the memory location that is allocated for containing some data such as a character string or an...' is 'array'.
5. True or False: The correct answer to 'The _____ is anything which your search engine cannot search.' is 'Deep web'.
6. True or False: The correct answer to 'Why apps developed in languages like C, C++ is prone to Buffer-overflow?' is 'Improper memory allocation'.
7. True or False: The correct answer to 'How does a buffer overflow on the stack facilitate running attacker-injected code?' is 'By overwriting the return address'.
8. True or False: The correct answer to 'A/an _____ is a program that steals your logins & passwords for instant messaging applications.' is 'Keylogger'.
9. True or False: The correct answer to 'A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.' is 'exploit'.
10. True or False: The correct answer to 'Let $I = (S,V)$ be a MAC. Suppose $S(k,m)$ is always 5 bits long. Can this MAC be secure?' is 'No, it cannot be secure'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to check whether the given number is prime or not.
2. Write a function to sort a tuple by its string element.

End of Paper